Chest-PRESS: Fine-tuned General vs. Medical LVLMs for Patient-Conditioned Chest X-Ray Report Generation

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Abstract

Building a chest X-ray report generator requires two choices: which backbone to fine-tune, and how much patient context to give it. Practice answers the first with zero-shot benchmarks and the assumption that medical pre-training beats general-purpose models, untested under controlled conditions because prior comparisons adapt one backbone family, confounding domain with architecture. In Chest-PRESS we fine-tune seven LVLMs, three general-purpose and four medical, with one identical LoRA recipe on 63,763 patient-conditioned MIMIC-CXR examples and evaluate them under one protocol with paired bootstrap intervals, six cleanly measured and one excluded for a serving defect. Both backbone-selection heuristics fail. Zero-shot rankings do not survive fine-tuning: the best finishes 4 of six, the weakest second. Domain pre-training yields no decisive advantage: the one architecturally controlled pair, identical but for pre-training data, differs by 0.002 RadGraph-F1, though the medical family holds a small lead on 14 of 14 CheXpert findings. It is the domain label that carries little information, not the backbone: backbones still span 0.094 RadGraph-F1 on FINDINGS. Prior reports carry nearly all the value of conditioning (+0.029 where they exist), while demographics, laboratory results and microbiology add +0.002. Backbone selection should rest on fine-tuned, architecture-controlled comparisons, not zero-shot scores or domain labels.

1 Introduction

Chest radiography is the highest-volume imaging examination in modern medicine, and every examination ends in a written report that turns a greyscale image into the findings and impression a physician acts on. Reading is the bottleneck: acquisition has become fast and cheap, reading has not, and reading lists grow faster than radiologist headcount.

Large vision-language models (LVLMs) offer an appealing intervention: image and clinical context in, FINDINGS and IMPRESSION out, and the radiologist corrects and signs a draft rather than composing one. But vision-language alignment in these models remains weak relative to their language priors, so a model can produce a fluent, plausible, well-formatted report that is not grounded in the image in front of it (Zhu et al., 2025). That is what makes the economics fail: a radiologist who cannot rely on any individual sentence must verify all of them, and verifying a fluent wrong sentence is harder than composing a correct one. The workload moves rather than shrinks.

Two responses have developed in parallel: richer input, conditioning on the patient's prior reports and indication, or a more medical model, through domain pre-training (Section 2). Whether the second still matters once the first is in place and the model is fine-tuned for the task is unestablished, because work here typically adapts one backbone family, entangling any general-versus-medical difference with an architecture difference. Nor has the upper end of conditioning been tested: structured electronic-health-record signal such as laboratory results and microbiology cultures, which a radiologist would certainly use, has neither been supplied to a report generator nor had its effect measured.

We run that comparison under controlled conditions (Figure 1), and find that the domain label largely stops mattering once the task is trained: after an identical LoRA recipe, whether a backbone was pre-trained on medical data explains less than its architecture does, and the one pair isolating domain from architecture is not separated at all.

Contributions. [1]

[1] Code, dataset-construction scripts, training and evaluation configuration, and the generators that produce every number and table in this paper: https://github.com/lks-hrsch/chest-press-cxr. The MIMIC data itself is not redistributable and must be obtained from PhysioNet under its

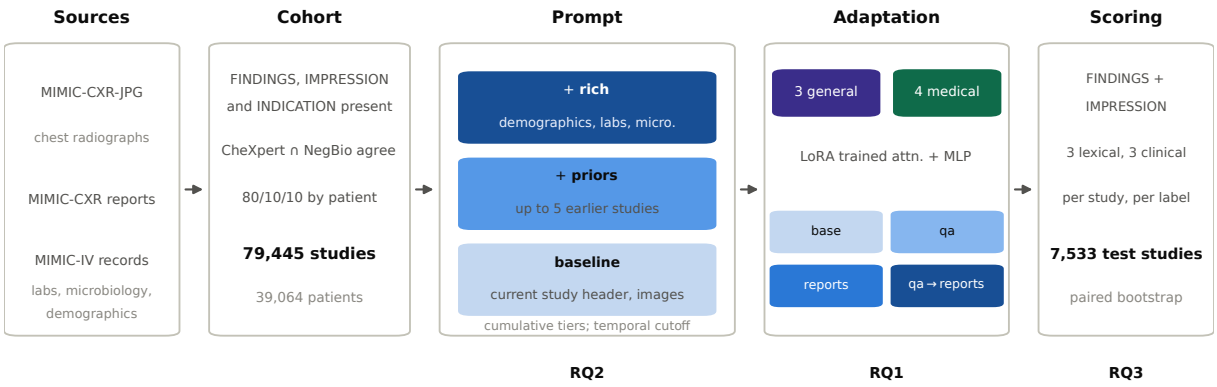


Figure 1: The study end to end. Counts are the full dataset and the test split, of which 7,533 studies are scored for every backbone, every training variant and every context tier. The three prompt blocks are cumulative, so `baseline` is the current study alone, `clinical_context` adds prior studies and `rich_clinical_context` adds demographics, laboratory and microbiology summaries.

- A **controlled comparison of seven LVLMs**, six of them cleanly measured, holding the adaptation recipe, the training data, the prompt and the evaluation protocol fixed across four training variants (`base`, `qa`, `reports`, `qa_reports`).
- A **patient-conditioned dataset and prompt** built from MIMIC-IV and MIMIC-CXR supplying prior studies, demographics and LLM-summarised laboratory and microbiology events, with a three-tier ablation isolating the **contribution of each block of clinical context**.

Research questions.

- **(RQ1)** Does LoRA fine-tuning of general-purpose LVLMs match or beat fine-tuning medical-domain LVLMs for chest X-ray report generation?
- **(RQ2)** How does rich patient clinical context (priors, indication, demographics, laboratory results, microbiology) influence the generated report?
- **(RQ3)** How do the models differ across finding types, and does the aggregate ranking hold within them?

2 Related Work

Clinically conditioned report generation. Early work treated report generation as image captioning, mapping a radiograph directly to free text. The consistent finding since then is that the image alone underdetermines the report: a radiologist writes with the indication, the prior study and the patient's history in view, and models given the same context produce more factually correct reports than image-only baselines (Kang et al., 2025). The prior study is particularly load-bearing, because much report content is explicitly differential ("unchanged", "improved", "new since"). We extend the conditioning further up the record, to demographics and structured laboratory and microbiology events, and measure whether that extension is worth its cost.

Medical-domain vision-language models. A parallel line of work adapts general LVLM recipes to biomedicine. LLaVA-Med (Li et al., 2023) applies the LLaVA (Liu et al., 2023, 2024a) instruction-tuning recipe to biomedical figure-caption data. MedGemma (Sellergren et al., 2025, 2026) continues Gemma (Gemma Team, 2025, 2026) pre-training on medical imaging including chest radiographs, and Lingshu (Xu et al., 2025) is a generalist medical multimodal model on an InternVL architecture. Each is released as an improvement over its general-purpose counterpart in the zero-shot or lightly adapted regime. What is typically not reported is how much of that advantage survives task-specific fine-tuning on in-domain data, and the comparisons that do exist rarely hold architecture fixed. Large-scale chest X-ray VQA corpora with quality-rated answers (Anonymous, 2025) additionally make an intermediate grounding stage possible, which we treat as an explicit training variant rather than an assumption.

Data Use Agreement.

Evaluating generated reports. Lexical metrics, BLEU (Papineni et al., 2002), ROUGE-L (Lin, 2004) and BERTScore (Zhang et al., 2020), measure how a report reads and are well known to be only loosely coupled to whether it is right: a report can score highly by reproducing the boilerplate that dominates radiology prose while inverting a single critical negation. Clinical metrics instead score the extracted content. RadGraph (Jain et al., 2021) extracts clinical entities and their relations and RadGraph-F1 (Delbrouck et al., 2022) compares the extracted graphs. The SRRG line of work (Delbrouck et al., 2025) scores structured reports against a multi-label finding taxonomy. We report both families, and as Section 5 shows they do not always agree on which model is best.

3 Approach

Task. We compare the seven backbones of Table 1, spanning four architecture families and 4.3–7.9 B parameters. Given the chest radiographs of a single study and a rendered clinical context, the model produces a FINDINGS section followed by an IMPRESSION section, in that order and with no others. This contract is fixed across all models and variants. The two sections are scored separately and in concatenation, because they are different tasks in practice: FINDINGS is descriptive and enumerative, IMPRESSION a synthesis.

Patient-conditioned prompt. Each example renders to one text prompt with three blocks in a fixed order (literal templates in Appendix A).

- **## Patient Information** carries sex, age and free-text summaries of the most recent laboratory and microbiology results
- **## Clinical Context** carries the patient's prior studies, oldest first, each with its own indication, findings and impression
- **## Current Study** carries the metadata of the study being reported (history, indication, examination, comparison, technique), wherever present in the source report

Laboratory and microbiology values are summarised into clinical prose by a separate LLM pass during dataset construction rather than passed as raw numbers, because raw event tables neither fit the context budget nor read like the clinical text these models were trained on.

Context ablation. RQ2 is operationalised as three tiers, each dropping a block from the unchanged rendered prompt, with no reordering and no regeneration:

- `rich_clinical_context` is the full prompt
- `clinical_context` removes `## Patient Information`
- `baseline` additionally removes the priors block, leaving only the current study

The two upper tiers share an identical system prompt, so the ablation isolates input content alone. Only `baseline` receives an edited prompt, because the shared wording refers to prior studies that tier no longer supplies.

Adaptation recipe. Every model is adapted with the same LoRA configuration (Hu et al., 2022): rank 32, α 64, dropout 0.1, on the attention and MLP projections of the language model only, with per-backbone details in Appendix B. These, the Adam optimiser and the cosine schedule follow Kang et al. (2025)'s Table 2, except the peak learning rate, halved to $1e{-4}$ because $2e{-4}$ re-invited the bf16 overflow that diverged `gemma-3-4b-it`. **The vision tower is frozen in every run.** The only per-backbone variation is mechanical: the attention implementation, and a micro-batch and accumulation split chosen to fit memory (Table 5). No hyperparameter is tuned per model for quality.

Training variants. Four variants per backbone. `base` is the released checkpoint, zero-shot. `qa` is a fresh LoRA adapter trained on chest X-ray VQA data, intended to establish visual grounding without teaching report style. `reports` is a fresh adapter trained on our report data, and `qa_reports` continues the trained `qa` adapter on that data as trainable LoRA weights – a continuation, not a merge. For inference, adapters are merged into base weights and served as full checkpoints, so all four variants share one serving path.

Model	Domain	Params	Year	Language model / vision encoder
`gemma-3-4b-it` (Gemma Team, 2025)	general	4.3 B	2025	Gemma 3 / SigLIP-400m
`medgemma-4b-it` (Sellergren et al., 2025)	medical	4.3 B	2025	Gemma 3 / SigLIP (med.)
`medgemma-1.5-4b-it` (Sellergren et al., 2026)	medical	4.3 B	2026	Gemma 3 / SigLIP (med.)
`gemma-4-E2B-it` (Gemma Team, 2026)	general	5.1 B	2026	Gemma 4 / MobileNet-V5
`llava-v1.6-mistral-7b` (Liu et al., 2024b,a, 2023)	general	7.6 B	2024	Mistral-7B / CLIP-336
`llava-med-v1.5-mistral-7b` (Li et al., 2023)	medical	7.6 B	2023	Mistral-7B / CLIP-336
`Lingshu-I-8B`‡ (Xu et al., 2025)	medical	7.9 B	2025	Qwen2-7B / InternViT

Table 1: The seven backbones: three general-purpose and four medical-domain, spanning four architecture families (separated by horizontal rules). ‡ `Lingshu-I-8B` is **excluded from the main comparison** for a serving defect and reported in Appendix C only. See Limitations.

4 Experimental Setup

Data. The report dataset is built from MIMIC-CXR (Johnson et al., 2019a, 2024c), MIMIC-CXR-JPG (Johnson et al., 2019b, 2024b) and MIMIC-IV (Johnson et al., 2023, 2024a). A study is retained only if it carries FINDINGS, IMPRESSION and INDICATION headings and if the CheXpert (Irvin et al., 2019) and NegBio (Peng et al., 2017) label sets agree on it, disagreement serving as a proxy for an ambiguous report. The result is 79,445 examples over 39,064 patients, split 80/10/10 by `subject_id` so no patient appears in two splits. The test split holds 7,533 distinct studies, 68.1% of them multi-view. Priors are available for 48.7% of them, which matters for RQ2 (Section 5.2).

Temporal cutoff. Every prompt block is restricted to information that existed before the target report was written, so no input can carry knowledge of the report the model must produce. Priors are the at most five earlier studies of the same patient, each contributing only its INDICATION, FINDINGS and IMPRESSION. Laboratory and microbiology events are the most recent hundred whose `storetime`, the moment a result became visible rather than the moment the specimen was taken, is at or before the current study: the conservative choice, since a test drawn before the examination but resulted after it is then correctly excluded. The summarisation pass sees only those events and the current indication, never the target report. Appendix D gives the rule per block, its enforcement and two caveats.

Training and inference. The `qa` variant trains for one epoch, with no evaluation split, on the highest-quality tier (A++) of MIMIC-Ext-CXR-QBA (Anonymous, 2025), whose one-million-question subset joins to 1,110,508 question-image pairs over 193,726 studies. The `reports` and `qa_reports` variants run 1 epoch over the training split, evaluating on the full validation split ten times per epoch and restoring the best checkpoint by validation loss. Both stages run on 4×H100 with Distributed Data Parallel and gradient checkpointing, truncating at 20,480 tokens. Generation runs in bf16 through vLLM with the same context window, at most 1,024 new tokens, and identical decoding parameters across models. Each backbone is served at its released image-processor configuration rather than a common resolution (Appendix B), a confound noted in the Limitations.

Metrics and uncertainty. We report the six metrics of Table 2, three lexical and three clinical, introduced in Section 2, each computed for FINDINGS, for IMPRESSION and for the two concatenated, except where one is undefined for a task (Table 10, Appendix G). RadGraph-F1 is our primary metric: it scores extracted clinical content rather than surface form and is defined on all three tasks. CheXbert-F1, conventionally reported alongside it, is not included, an omission revisited in the Limitations. Differences between backbones here are small, so every comparison the paper argues from carries a paired bootstrap 95% confidence interval in brackets (Efron and Tibshirani, 1993; Koehn, 2004). Because both systems are evaluated on the same studies, we resample *studies* with replacement and recompute the mean per-study difference over 10,000 replicates, which removes the between-study difficulty variation that would otherwise dominate the interval (Appendix E). **These intervals quantify uncertainty from the finite test set only.** Every model was trained once, so a difference whose interval excludes zero is established over studies rather than over training runs.

Backbones	6 in the main comparison, 7 trained (Table 1)
Training data	63,763 patient-conditioned MIMIC-CXR studies
Validation / test	7,822 / 7,860 rows, split by `subject_id`
Adaptation	LoRA r=32, α=64, dropout 0.1, attention + MLP, vision tower frozen
Training variants	`base`, `qa`, `reports`, `qa_reports`
Context tiers	`baseline`, + priors, + rich (demographics, labs, microbiology)
Metrics	3 lexical (BLEU, ROUGE-L, BERTScore), 3 clinical (RadGraph-F1, F1-SRR-BERT, Category Score)
Primary metric	RadGraph-F1, defined on all three tasks
Scored studies	7,533 (all backbones, all variants, all tiers)
Uncertainty	paired percentile bootstrap over studies, one training seed
Hardware	4×H100 for training, 1×RTX 4090 for inference and scoring

Table 2: The experimental setup at a glance. Appendix B gives the per-backbone configuration, hyperparameters and compute.

5 Results

5.1 RQ1: general-purpose versus medical-domain backbones

Fine-tuning improves every backbone on every metric. The interesting effect is on the differences between them. Zero-shot, the six backbones span 0.268 RadGraph-F1 on `combined` (0.326 down to 0.059). After identical LoRA fine-tuning that spread collapses to 0.082, a 3.2× reduction (Figure 6, Appendix I). They converge, and not in rank order: the best zero-shot model, `medgemma-1.5`, finishes 4 of six, while the weakest, `llava-med-v1.5`, finishes second, ahead of every model that beat it before training.

There is no clean general-versus-medical split, and the two halves of that statement pull in opposite directions. The top two backbones, `llava-v1.6` (general) and `llava-med-v1.5` (medical), are separated by 0.002 RadGraph-F1 on FINDINGS with an interval spanning zero ($[-0.001, +0.005]$), while the gap from second to third is +0.017 $[+0.014, +0.021]$ and does not. That leading pair is a tier rather than a ranking, and it is also our one architecturally controlled comparison: same language model, same vision encoder, differing only in domain. Yet the medical family mean does exceed the general one (0.428 against 0.388, a paired +0.041 $[+0.038, +0.043]$, and Figure 8). Architecture tracks the outcome more closely than domain, the two Mistral-7B/CLIP models taking the top two positions regardless of which is medical. The two metric families disagree at the top of Table 3 too: `llava-v1.6` wins all three lexical metrics and RadGraph-F1, but F1-SRR-BERT and Category Score both go to `medgemma-4b`, which wins none of the other four. Which model is "best" depends on which clinical metric is asked, a point we return to in Section 6.

Training variants. The `qa`-only variant is not merely unhelpful but destructive, averaging 0.076 RadGraph-F1 and falling below untrained `base` on 5 of 6 backbones (Appendix I, discussed in Section 6). Table 14 in that appendix shows what it looks like on one study. The `qa_reports` curriculum is small but measurable and inconsistent in sign: 4 of 6 per-backbone intervals exclude zero, 3 of them degradations, and none moves a backbone by more than 0.021 (Table 8, with all four variants in Figure 9 and per-metric tables in Tables 12 and 13). It is as likely to hurt as to help, and whether that survives a second seed is unquantified.

5.2 RQ2: how much clinical context is worth supplying

Table 11 and Figure 7 (Appendix I) give the three-tier ablation per backbone. Relative to the `baseline` tier, adding prior studies raises RadGraph-F1 by +0.015 $[+0.013, +0.017]$ on fine-tuned models, and additionally adding demographics, laboratory results and microbiology raises it by a further +0.002 $[+0.001, +0.003]$. The direction is consistent across all backbones (+0.012 to +0.023 baseline-to-rich). Both intervals exclude zero, so neither block is inert. What separates them is size: prior reports are worth about 8 times what the structured block adds. That block is a marginal return on the most expensive component of our dataset construction.

The structured block does not improve when it carries more. If that small effect were an averaging artifact, most studies having nothing to report, then restricting to substantive summaries should recover a larger gain. It does not: the block is worth +0.002 where the laboratory summary says something against +0.001 where it does not, microbiology behaves the same way, and infection-

	Lexical			Clinical		
Model	BLEU	ROUGE-L	BERTScore	RadGraph-F1	F1-SRR-BERT	Cat. Score
`llava-v1.6`	**0.333**	**0.483**	**0.917**	**0.446**	0.681	0.781
`llava-med-v1.5`	0.332	0.482	0.917	0.445	0.678	0.782
`medgemma-4b`	0.306	0.459	0.914	0.427	**0.684**	**0.791**
`medgemma-1.5`	0.293	0.448	0.912	0.414	0.674	0.785
`gemma-3-4b`	0.242	0.404	0.903	0.364	0.636	0.731
`gemma-4-E2B`	0.232	0.396	0.902	0.353	0.630	0.726

Table 3: Cross-model results on FINDINGS: `reports` variant, over 7,531 scored test studies. Best per column **underlined**.

	Lexical			Clinical	
Model	BLEU	ROUGE-L	BERTScore	RadGraph-F1	F1-SRR-BERT
`llava-v1.6`	**0.317**	0.426	0.910	0.382	0.604
`llava-med-v1.5`	0.315	**0.427**	**0.910**	**0.383**	0.606
`medgemma-4b`	0.300	0.425	0.909	0.382	0.617
`medgemma-1.5`	0.293	0.415	0.909	0.373	**0.619**
`gemma-3-4b`	0.283	0.391	0.894	0.352	0.578
`gemma-4-E2B`	0.280	0.391	0.894	0.353	0.578

Table 4: Cross-model results on IMPRESSION: `reports` variant, over 7,531 scored test studies. Best per column **underlined**.

related studies do not help either (Table 9, Appendix F). It is weak where it should be strongest, so this is not dilution. Before fine-tuning the pattern even reverses, the richest tier scoring *worse* than the priors-only one (0.185 against 0.193, -0.008 $[-0.009, -0.007]$). Untrained models are distracted by context they have not learned to use, whereas fine-tuned models learn to ignore it, so the benefit of conditioning is partly a benefit of having been trained to condition.

5.3 RQ3: where the models differ

Every metric is stored per study with its CheXpert labels, and the aggregate ranking does not hold uniformly within them.

By finding type. Restricting to studies positive for each CheXpert label (Figure 2, Appendix H), the best model changes with the label: `medgemma-4b` wins 7 of the fourteen and `llava-v1.6` 6, the fourteenth (Cardiomegaly) going to `medgemma-1.5`, and no other backbone wins one. Underneath that alternation is a more consistent pattern: the medical family mean leads on 14 of 14 labels, by +0.013 to +0.045 RadGraph-F1. Unlike the controlled pair, the per-label picture is one-sided. Absolute scores also vary far more across labels than across models, `No Finding` reaching 0.566 while the median pathology label reaches only 0.301. Reporting a normal study is much easier than characterising an abnormality, and an aggregate metric on a cohort where 51.6% of studies are labelled `No Finding` partly measures the former.

By report section. The FINDINGS leader does not carry over. On FINDINGS `llava-v1.6` leads RadGraph-F1 clearly. On IMPRESSION (Table 4) the top three backbones fall within 0.001 of one another, so no leader can honestly be named, and the per-label picture is correspondingly flat (Figures 4 and 5). IMPRESSION is also far more compressed, the six backbones spanning 0.031 RadGraph-F1 against 0.094 on FINDINGS, so most of the between-model difference this paper measures is a difference in how the models write FINDINGS. Describing the image separates our backbones. Synthesising a one-line impression barely does.

6 Analysis, Conclusion and Future Work

The central result is not a ranking of models. It is that once the task is trained, medical multimodal

pre-training provides little aggregate advantage, while substantial differences remain across architectures. Four measurements support and qualify it, in order of weight.

The architecturally controlled pair. `llava-v1.6` and `llava-med-v1.5` share a language model and a vision encoder and differ only in pre-training domain. After identical fine-tuning the interval on their difference covers zero (Section 5.1), the only place in the study where domain is isolated from architecture. *The per-finding breakdown.* It would be wrong to conclude from that pair that domain pre-training does nothing: the medical family mean leads on 14 of 14 CheXpert labels (Section 5.3) and the family means differ by +0.041 $[+0.038, +0.043]$. It leaves a small, broad residue, visible averaged over many findings but not decisive between two otherwise identical models. *The rank inversion.* The best zero-shot backbone finishes 4 of six and the weakest second, and the spread compresses 3.2-fold, so what a zero-shot benchmark measures is largely not what survives adaptation. *The metric disagreement.* No single backbone wins both metric families on FINDINGS (Table 3): on a task where the models are this close, the choice of clinical metric moves the answer as much as the choice of backbone.

Architecture, the adaptation recipe, prior-report conditioning and the chosen metric therefore each explain more of the variation than the general-versus-medical label does. That is also RQ3's answer from the other direction: the aggregate ranking hides a per-subgroup picture in which the best model depends on the finding type and the report section.

Why VQA-only training is destructive. That `qa` scores below `base` is surprising, given that the VQA data is in-domain chest radiography with quality-rated answers. The explanation is that Stage A teaches a format, not just a grounding: a model trained on a million questions to answer in one sentence learns to emit a sentence, and asked for a two-section report emits something question-answer-shaped that the report metrics score accordingly (Table 14, Appendix I). The `qa_reports` curriculum agrees: once a `reports` stage follows, the format damage is repaired and only a backbone-dependent effect of at most 0.021 remains (Section 5.1). Whatever grounding `qa` confers does not survive in a form the report metrics can see, so under this recipe an intermediate VQA stage is not worth its compute.

RQ2: what clinical conditioning is worth, and why. The pooled +0.015 for priors understates them, because the block cannot help a study that has none and only 48.7% of test studies have any. Conditioning on whether a study has priors recovers +0.029 $[+0.025, +0.033]$, against +0.001 where it has none, a null-manipulation control that is not quite zero for a reason traced to the tier design in Appendix F. For a returning patient, priors are worth roughly twice the headline figure. Much of that is likely finding persistence rather than reasoning over history: across the 3,626 test studies with a matchable earlier study, the positive CheXpert label set is *identical* to the previous one in 30.2% of cases and agrees per label 88.0% of the time, so a model handed the previous report can carry a finding forward and usually be right. A radiologist uses the same regularity, but it is a weaker claim than understanding a history, and separating the two needs a manipulation holding persistence fixed, which we did not run.

6.1 Practical recommendations

Read as a design guide, the study settles two decisions any project adapting an LVLM to a medical reporting task has to make.

1. **Choose the backbone by fine-tuned, architecture-controlled evidence, not by zero-shot score or by domain label.** The rank inversion above leaves a zero-shot leaderboard close to uninformative about the fine-tuned ordering, and the "medical" label is no more useful for an individual model: the best *and* the weakest fine-tuned backbone here are both general-purpose.

2. **Build prior-report conditioning, and skip the structured clinical summaries.** The conditioned priors effect above is an order of magnitude larger than the +0.002 the structured block adds, and that block is by far the most expensive part of dataset construction, a multi-GPU summarisation pass over the whole cohort.

Future work. **Staged generation**: the FINDINGS leader does not carry over to IMPRESSION, so a two-model pipeline is worth measuring against the best single one. **Better structured-data conditioning**: the weak RQ2 effect bounds the value of

our LLM-produced summaries, not of the underlying signal. **Training budget**: one seed at 1 epoch and one unsearched learning rate could plausibly reorder a ranking this compressed. **Prompting**: chain-of-thought reasoning (Wei et al., 2022) fits a task that is explicitly observe-then-synthesise.

Limitations

One backbone is excluded for a serving defect. Every `Lingshu-I-8B` figure was produced through a vLLM language-model path that corrupts output for this checkpoint (Appendix C): served zero-shot, it emits non-Latin characters in place of a report. Its scores are a lower bound of unknown tightness on a broken serving path rather than a measure of model quality, so it is excluded from Sections 5 and 6 and reported in the appendix for transparency. No claim here depends on it.

Empty generations are a scoring boundary case. Zero-shot `llava-med-v1.5` returns empty output for 46.6%/5.6%/10.3% of studies across the three context tiers (non-monotonic), scored as zero rather than excluded – a convention making its zero-shot numbers not directly comparable to figures reported elsewhere. Fine-tuning removes the issue (0.0% blank), so no fine-tuned number is affected.

Architecture is confounded with scale. Backbones are compared as released, so architecture family co-varies with parameter count, visual token budget and effective batch size. Two of the three are bounded by figures already reported: the token budget differs fivefold inside the controlled pair (Appendix B) without separating it, and the leading backbones are the ones trained at the smaller effective batch (Table 5). Parameter count is not separable, since the two leading backbones are also the largest cleanly measured ones. Family-level statements here should therefore be read as architecture-or-scale. Only the controlled pair isolates pre-training domain.

One recipe, one dataset, one fine-tuning-set size. Every result is measured at 63,763 training examples, with one LoRA configuration, on one dataset and one task. The finding that domain pre-training matters little after fine-tuning is therefore a statement about this regime, not about adaptation in general: a medical checkpoint starts closer to the target, so its advantage would be expected to grow as the fine-tuning set shrinks, and nothing here bounds where that crossover lies. In-context learning is also untested, since we compare zero-shot and fine-tuned models only.

No radiologist evaluation. Every claim here rests on automatic metrics. The clinical ones score extracted content and are better proxies than lexical overlap, but no radiologist read any generated report, and the qualitative examples involve manual clinical judgement by the author, who is not a clinician.

Label-conditioned analysis inherits automatic-labeller error. The RQ3 breakdown groups studies by CheXpert labels, themselves derived automatically from the reference reports. Our agreement filter reduces but does not eliminate label noise, and several labels have small support, so per-label rankings on the rarest findings rest on few studies. The fourteen subgroups also overlap heavily, so the family gap holding on all fourteen is fewer than fourteen independent observations, and how much of it reflects common disease combinations rather than per-finding effects our analysis does not separate.

Data use. All imaging, reports and clinical data are the de-identified MIMIC-CXR/MIMIC-IV releases, used under their PhysioNet Data Use Agreement: no image, report or derived text left the credentialed environment, and the laboratory/microbiology summarisation pass (Section 4) ran a locally hosted model rather than a third-party API.

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Generative AI disclosure. We used generative AI tools for code refactoring, documentation, boilerplate scripts and language polishing of this manuscript. They did not contribute to research ideation, experimental design or analysis: the experiments, the analysis and the ideas are the authors' own, and every reported number was produced by the described pipeline and verified by the authors.

[2] https://faerber-lab.github.io/team/, https://scads.ai/about-us/ai-professorships/scalable-software-architectures-for-data-analytics/

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A Prompt templates

Section 3 describes the three-block structure and the context ablation. This appendix gives the literal templates, so that the ablation is reproducible without reading the generator source. All quoted report text below is drawn from the de-identified MIMIC-CXR release, kept to the minimal excerpt needed, and reproduced verbatim, including its own irregularities (abbreviations, an occasional typo, and the ___ de-identification placeholders).

System prompt. `rich_clinical_context` and `clinical_context` share one system prompt, since both tiers genuinely carry clinical context and the ablation should isolate input content, not prompt wording:

> You are an expert radiologist. You are given some clinical context about a patient and one or two chest X-Ray images from the patient in different views. Write the IMPRESSION and FINDINGS sections of the report. Consider the available clinical context about the patient when generating the report.

Only `baseline` receives an edited prompt, because the wording above refers to context that tier does not supply:

> You are an expert radiologist. You are given a chest X-Ray study (one or two images in different views) along with the current study's exam details (indication, technique, comparison). Write the IMPRESSION and FINDINGS sections of the report.

Rendered example, all three tiers. One test-split study (de-identified, MIMIC-CXR), shown at each tier. `baseline` is the last block only, `clinical_context` adds the prior-study block, and `rich_clinical_context` additionally adds patient information. No block is reordered or regenerated between tiers.

> **baseline / clinical_context / rich_clinical_context – current study (all three tiers):**
> `## Current Study (2131-09-12 11:39:50):`
> Clinical Context (last report is most recent):
> INDICATION: History: ___F with CP, recvent URI // r/o PNA
> EXAMINATION: Chest: Frontal and lateral views
> COMPARISON: None.
> TECHNIQUE: Chest: Frontal and Lateral
>
> **clinical_context / rich_clinical_context add – prior study block:**
> `## Clinical Context (last report is most recent):`
> `### 2130-04-06 17:29:59`
> INDICATION: Persistent cough. FINDINGS: The heart is normal in size. The mediastinal and hilar contours appear within normal limits. There is no pleural effusion or pneumothorax. The lungs appear clear. IMPRESSION: No evidence of acute cardiopulmonary disease.
>
> **rich_clinical_context adds – patient information block:**
> `## Patient Information`
> GENDER: F
> AGE: 27
> LAST_LAB_FINDINGS: No significant findings.
> LAST_MICROBIOLOGY_FINDINGS: No significant findings.

`rich_clinical_context` is these three blocks concatenated in the order patient information, prior studies, current study. `clinical_context` drops the first, `baseline` keeps only the last. The target this study's models are trained to produce (`OUTPUT_LABELS`, FINDINGS then IMPRESSION, no other section) is:

> FINDINGS: The lungs are clear without focal consolidation. No pleural effusion or pneumothorax is seen. The cardiac and mediastinal silhouettes are unremarkable. IMPRESSION: No acute cardiopulmonary process.

Laboratory and microbiology findings, when present, are not raw event tables: a separate LLM pass (`google/gemma-4-26B-A4B-it` via vLLM, during dataset construction) summarises them into the one or two clinical-prose sentences that appear after `LAST_LAB_FINDINGS` and `LAST_MICROBIOLOGY_FINDINGS` above. This study's summaries default to "No significant findings." because it has no recorded events in the lookback window.

B Full setup: hyperparameters, serving and compute

Table 5 gives the per-backbone training configuration. All values are read directly from `training_config / sft_batch` in `src/model_registry.py`. LoRA target modules are identical across every backbone and are not repeated per row: rank 32, α 64, dropout 0.1, on `q_proj/k_proj/v_proj/o_proj` and `gate_proj/up_proj/down_proj` of the language-model backbone only (Section 3).

Attention implementation. The Gemma family uses SDPA rather than FlashAttention-2: FA2 cannot consume Gemma 3's additive bidirectional image-attention mask, so training would either error or silently fall back to a slower path. SDPA has

Backbone	Attn.	Liger	Eff. batch
gemma-3-4b	SDPA	N	64
gemma-4-E2B	SDPA	N	64
medgemma-4b	SDPA	N	64
medgemma-1.5	SDPA	N	64
llava-med-v1.5	FA2	Y	32
llava-v1.6	FA2	Y	32
Lingshu-I-8B	FA2	Y	32

Table 5: Per-backbone training configuration. Effective batch = per-device batch × 4 GPUs × gradient-accumulation steps, held constant per model across the qba and reports stages even where the per-device/accumulation split differs between them (e.g. Gemma: 16×1 on qba, 8×2 on reports, both effective 64) to fit the heavier multi-image reports batch in memory.

no such restriction and was verified to train Gemma-3 with gradient checkpointing at the batch sizes in Table 5. LLaVA and Lingshu use FA2, which is the faster, lower-memory choice at their larger per-image token counts (up to 2,880 for LLaVA-Next's AnyRes tiling) once the mask restriction does not apply.

Liger kernel. Disabled specifically for the Gemma family: liger-kernel 0.8.0's Gemma-3 vision-tower patch dereferences `vision_tower.vision_model.post_layernorm`, an attribute path that transformers 5.x removed by hoisting `post_layernorm` directly onto `SiglipVisionModel` – a fatal `AttributeError` before the first training step, not a performance regression. Setting `layer_norm: False` in Liger's own config does not avoid it, because that flag only gates the encoder-internal layer norms, not this top-level access. Enabled on every other backbone.

One backbone needs an excluded LoRA region. `gemma-4-E2B-it` (Gemma 3n) wraps its vision and audio encoders' `q_proj/k_proj/v_proj/o_proj/gate_proj/up_proj/down_proj` projections in a custom `Gemma4ClippableLinear` layer that PEFT cannot inject LoRA into. Left unhandled, PEFT's target-module name-matching would attempt it anyway and fail. The registry instead excludes both towers and their embedding bridges by name pattern before target matching runs, so LoRA lands only on the language-model projections in Table 5's row – the one place in the registry where "the vision tower is frozen in every run" (Section 3) needed an explicit carve-out rather than following from the target-module list alone.

A wrong attribute name would have unfrozen Lingshu's vision tower. Every backbone's vision encoder is frozen by module attribute name before training. By analogy with the other architectures this was first guessed as `.visual` or `.vision_model`. Lingshu-I-8B loads as transformers' `InternVLForConditionalGeneration`, whose encoder in fact lives at `model.model.vision_tower`. The wrong attribute would have left the tower unfrozen and, because Lingshu's InternViT layers also expose `q_proj/k_proj/v_proj`, would have let LoRA inject into the vision encoder as well, since the derived exclude-pattern matched nothing under the wrong name. Caught and corrected before any full training run. The value in Table 5 is the fixed one, and happens to coincide with every other backbone's attribute name once corrected.

Lingshu's tiling is the one non-default vision-processor override. `mm_processor_kwargs = {"max_dynamic_patch": 4}` caps InternVL's dynamic image tiling at 4 patches (∼1,280 tokens/image at 4 images), matching the token budget used throughout. This is a separate issue from the vLLM serving-path corruption documented in Appendix C: the value is expressed in vLLM's own vocabulary, and for training (which serves the checkpoint through transformers, not vLLM) it has to be translated to that processor's `crop_to_patches/max_patches` keys. An earlier version passed the vLLM-shaped dict straight through, and transformers silently ignored the unrecognised key rather than erroring, serving 13 tiles (∼3,335 tokens/image) instead of the intended 4 (∼1,287), a 2.6× prefill and KV-cache cost inflation with no error to surface it. The translation is centralised in one helper with a fail-fast assertion that turns a future ignored processor kwarg into a build error instead of a silent 2.6× regression.

Training and generation cadence. Beyond the cadence given in Section 4, Stage A (qa) checkpoints ten times per epoch. Truncation happens in the collator and keeps the start of the sequence, since truncating from the left can drop image placeholder tokens and desynchronise the visual features from the token sequence. Keep-start is the tokeniser default here rather than an explicit setting. Generation serves every image in the study, at most five in the test split, and the four-image cap applies to training only.

Why visual input resolution is not equalised. Each model's image processor is configured to its native resolution and tile budget, so the per-image token cost ranges from 256 (the SigLIP-based Gemma models) to as much as 2,880 (LLaVA-Next with AnyRes). Equalising this is not possible without handicapping some backbones: only the InternVL-style encoder decouples token count from input resolution, whereas LLaVA-Next's AnyRes couples the two, so a common token budget would also be a common and lower resolution. Since MIMIC-CXR-JPG images are natively about 2544×3056, every encoder here already downsamples heavily, and serving each model at its released configuration compares the models as they are actually deployed rather than at an artificial common setting.

An adapted metric. F1-SRR-BERT and Category Score are applied here to free-text reports, by running the labeller per sentence and merging its predictions per report, then rolling the leaf labels up to organ categories. They were designed for the structured report format, whose organ headers the rollup substitutes for. The model, taxonomy and aggregation are the source implementation's, so the values are comparable in kind but not numerically to the papers that introduced them.

Compute. Training ran on $4 \times$H100. Inference and evaluation both ran on a single RTX 4090. The full grid is 7 models $\times$ 4 training variants for generation, plus the three-tier ablation: generation averages $\sim$40 min per model/variant/tier pass over the full 7,860-study test split. Separately, the dataset-construction pass took $\sim$12 h ($4 \times$H100), and the full evaluation re-score took $\sim$9.7 h wall-clock on the 4090.

C Lingshu serving-path diagnosis

This appendix is the evidence for the caveat stated in the Limitations section: that `Lingshu-I-8B`'s numbers were produced through a serving path that corrupts this checkpoint's output, and are therefore not a measurement of model quality.

The observation. Given the same checkpoint, the same prompt and the same images, the two runtimes disagree completely. Under transformers the model produces a well-formed report, opening *"Findings: The lungs are clear. There is no pleural effusion or pneumothorax… Impression: No acute pulmonary disease."* Under vLLM (the runtime used for every inference run in this paper) the same request returns degenerate text: repeated Latin fragments interleaved with CJK characters, with no report structure and no clinical content. At the `base` variant this yields a RadGraph-F1 of exactly 0.059, which is a parsing outcome rather than a score.

What we ruled out. Each of the following was tested and eliminated as the cause: the dynamic-tiling configuration, the chat template, the weight-name structure of the checkpoint, the `crop_to_patches` processor attribute, `rope_scaling`, `torch.compile` and CUDA-graph capture, the compute dtype, and the context length. The tiling check did turn up a separate real bug, in which the tiling keyword was silently dropped by the transformers-backed processor, fixed independently of this issue.

The vision path is not involved. The decisive test is that the corruption reproduces on a twelve-token, text-only prompt with no image attached. That excludes the image processor, the vision encoder, the projector and the visual-token budget in a single step, and localises the fault to vLLM's language-model path for this checkpoint. `Lingshu-I-8B` is the only InternVL-architecture model in our roster, and it is the only model affected. The other six agree between runtimes.

What this does and does not invalidate. It invalidates every Lingshu row, not only the visibly broken `base` one. The fine-tuned variants emit well-formed reports, but only because LoRA supplied a strong output prior: their image grounding went through the same corrupted path, so their scores are a lower bound of unknown tightness. This is why the backbone is excluded from the main comparison outright rather than carried with a footnote. Two consequences show what including it would have done. First, its zero-shot 0.000 sets the floor of the between-backbone spread, so the compression ratio reads $4.0\times$ with it against $3.2\times$ without. Second, and more seriously, it is one of four medical backbones, so it drags the medical family mean down: the medical-minus-general gap at `base` is -0.019 including it and 0.031 excluding it, a change of sign. An RQ1 family claim made on the seven-model set would therefore have pointed the opposite way for a reason that has nothing to do with medical pre-training.

The full seven-backbone tables. Tables 6 and 7 reproduce the two main results tables with `Lingshu-I-8B` restored, so the excluded rows are on the record. They are diagnostic, not evidential. The figures throughout this appendix are likewise drawn over all seven backbones.

Remedy, and why it was not applied. Serving `Lingshu-I-8B` through transformers produces correct output and would resolve this. It was not applied because it is not only a matter of spending more GPU time: our inference entry point is built on vLLM throughout, with generation issued as a single batched `chat` call, so a transformers-served backbone requires implementing and validating a second serving path, then regenerating and rescoring four training variants across the test split. That was a larger piece of work than the defect's effect on our conclusions justified, and it is the first item of unfinished business.

That effect is bounded, and in one direction the bound is reassuring. Restoring `Lingshu-I-8B` to the medical family lowers the family-mean gap on FINDINGS from +0.041 $[+0.038, +0.043]$ to +0.030 $[+0.028, +0.032]$ but does not reverse it, so the direction of the RQ1 family comparison does not depend on the exclusion. What depends on it is the size of an effect the paper already describes as small and broad. The individual scores remain uninterpretable, since a lower bound measured through a corrupted decoder is not a measurement of the model.

	Lexical			Clinical		
Model	BLEU	ROUGE-L	BERTScore	RadGraph-F1	F1-SRR-BERT	Cat. Score
`llava-v1.6`	**0.333**	**0.483**	**0.917**	**0.446**	0.681	0.781
`llava-med-v1.5`	0.332	0.482	0.917	0.445	0.678	0.782
`medgemma-4b`	0.306	0.459	0.914	0.427	**0.684**	0.791
`medgemma-1.5`	0.293	0.448	0.912	0.414	0.674	0.785
`Lingshu-I-8B`	0.280	0.421	0.907	0.386	0.670	**0.794**
`gemma-3-4b`	0.242	0.404	0.903	0.364	0.636	0.731
`gemma-4-E2B`	0.232	0.396	0.902	0.353	0.630	0.726

Table 6: Table 3 with `Lingshu-I-8B` restored. Its row reflects the corrupted serving path diagnosed above and is reported for completeness only. Note that it takes the Category Score column, which is one of the results excluding it changes.

	Lexical			Clinical	
Model	BLEU	ROUGE-L	BERTScore	RadGraph-F1	F1-SRR-BERT
`llava-v1.6`	**0.317**	0.426	0.910	0.382	0.604
`llava-med-v1.5`	0.315	**0.427**	**0.910**	**0.383**	0.606
`medgemma-4b`	0.300	0.425	0.909	0.382	0.617
`medgemma-1.5`	0.293	0.415	0.909	0.373	**0.619**
`Lingshu-I-8B`	0.283	0.405	0.907	0.360	0.604
`gemma-3-4b`	0.283	0.391	0.894	0.352	0.578
`gemma-4-E2B`	0.280	0.391	0.894	0.353	0.578

Table 7: Table 4 with `Lingshu-I-8B` restored, on the same terms as Table 6.

D Temporal cutoff of the clinical context

Every input block is restricted to information that existed before the target report was written. The rule per block:

- **Prior studies**: the at most five studies of the same patient whose acquisition timestamp is *strictly* earlier than the current study. Each contributes only its indication, findings and impression.
- **Laboratory events**: the most recent one hundred events with `storetime` $\leq$ the current study's timestamp.
- **Microbiology events**: the same rule and the same window.
- **Derived summaries**: the summarisation pass receives exactly those events plus the current study's indication. It never receives the target findings or impression, any later study, or any event past the cutoff, so the summaries inherit the cutoff of their inputs.
- **Demographics**: sex and age, patient-level constants.
- **Current-study block**: the header fields of the target report (history, indication, examination, technique, comparison).

The cutoff is enforced mechanically. Events are held in a per-patient index sorted by `storetime` and selected by binary search against the study timestamp, so no filter can be forgotten for a subset of rows. The timestamp normalisation this depends on is covered by the test suite, after a defect in which a pre-10:00 study time was right-padded rather than left-padded, inflating it so that it sorted after every real same-day event and admitted post-study laboratory results.

Two caveats we do not hide. First, the current-study block is taken from the same report document that supplies the target findings and impression. Those header fields are dictated before the images are read and state why the examination was ordered rather than what it showed, and conditioning on the indication is standard for this task. The field

to watch is `COMPARISON`, which can reveal that a prior exists, though not what it showed. Second, MIMIC-IV records age at a patient-level anchor year rather than at each study, so our age field is a patient constant and not the age on the day of the examination. Neither can transmit information created after the target report.

E Paired bootstrap confidence intervals

Definition. For two systems evaluated on the same n studies, let $d_i = x_i - y_i$ be the per-study difference in the metric and $\hat{\theta} = \frac{1}{n}\sum_i d_i$ the observed mean difference. For each of B replicates we draw an index vector $I^{(b)} = (i_1, \ldots, i_n)$ uniformly with replacement from $\{1, \ldots, n\}$ and compute $\hat{\theta}^{*(b)} = \frac{1}{n}\sum_j d_{i_j}$, applying the *same* index vector to both systems. The $1-\alpha$ interval is the pair of empirical quantiles $[\hat{\theta}^*_{(\alpha/2)}, \hat{\theta}^*_{(1-\alpha/2)}]$ of those replicates, that is the percentile method (Efron and Tibshirani, 1993). The paired form is the construction introduced for machine translation by Koehn (2004) and now standard in NLP evaluation.

Why paired, and why bootstrap. Per-study RadGraph-F1 is bounded, zero-inflated and right-skewed, so the normality assumption behind a t-interval does not hold, and the bootstrap assumes no parametric form. Pairing matters even more: study difficulty varies far more on this task than the models do (Section 5.3), and that variation is shared by both systems. Resampling each system independently would leave it in the comparison and inflate every interval, whereas resampling studies cancels it inside the difference. Because $\hat{\theta}$ is a mean of paired differences, resampling studies and resampling d directly are equivalent, which is why this is cheap. That equivalence holds for all six metrics here because each is stored and aggregated per study. A corpus-level statistic would have to be recomputed inside the loop.

Procedure. Both arms are filtered to one (model set, variant, task, context tier), reduced to one value per study, and inner-joined on `(subject_id, study_id)`. That join is what enforces pairing. Family-mean arms average across their backbones *per study* first, so the unit of resampling is always the study. We use $B = 10{,}000$, $\alpha = 0.05$ and a pinned seed, with an independent random stream per comparison so that adding one does not shift the others. Studies are sorted before resampling, so every interval below reproduces exactly on a rebuild, and a 2,000-replicate run agrees with the reported values to three decimals.

Comparison	n	$\hat{\theta}$	95% CI
Between backbones (`reports`*)*			
`llava-v1.6` − `llava-med-v1.5` (controlled pair)	7,531	$+0.0019$	$[-0.0008, +0.0046]^{\dagger}$
`llava-med-v1.5` − `medgemma-4b` (2nd − 3rd)	7,531	$+0.0173$	$[+0.0136, +0.0211]$
medical − general family mean (7 backbones)	7,531	$+0.0300$	$[+0.0279, +0.0321]$
medical − general family mean (6 backbones)	7,531	$+0.0407$	$[+0.0385, +0.0430]$
`qa_reports` − `reports`			
`llava-v1.6`	7,531	$+0.0005$	$[-0.0026, +0.0035]^{\dagger}$
`llava-med-v1.5`	7,531	$+0.0000$	$[-0.0030, +0.0031]^{\dagger}$
`medgemma-4b`	7,531	-0.0087	$[-0.0117, -0.0058]$
`medgemma-1.5`	7,531	-0.0117	$[-0.0150, -0.0083]$
`Lingshu-I-8B`	7,531	-0.0188	$[-0.0223, -0.0155]$
`gemma-3-4b`	7,531	$+0.0210$	$[+0.0179, +0.0241]$
`gemma-4-E2B`	7,531	-0.0066	$[-0.0098, -0.0034]$
Context tiers, pooled			
+ priors (`clinical_context` − `baseline`)	7,531	$+0.0145$	$[+0.0127, +0.0165]$
+ rich (`rich_clinical_context` − `clinical_context`)	7,531	$+0.0018$	$[+0.0009, +0.0027]$
total (`rich_clinical_context` − `baseline`)	7,531	$+0.0163$	$[+0.0143, +0.0183]$
zero-shot: + rich (base variant)	7,531	-0.0079	$[-0.0088, -0.0070]$
Context tiers, per backbone			
+ priors, `llava-v1.6`	7,531	$+0.0164$	$[+0.0137, +0.0189]$
+ priors, `llava-med-v1.5`	7,531	$+0.0168$	$[+0.0142, +0.0194]$
+ priors, `medgemma-4b`	7,531	$+0.0089$	$[+0.0062, +0.0117]$
+ priors, `medgemma-1.5`	7,531	$+0.0129$	$[+0.0101, +0.0158]$
+ priors, `Lingshu-I-8B`	7,531	$+0.0076$	$[+0.0048, +0.0105]$
+ priors, `gemma-3-4b`	7,531	$+0.0140$	$[+0.0112, +0.0169]$
+ priors, `gemma-4-E2B`	7,531	$+0.0183$	$[+0.0151, +0.0216]$
+ rich, `llava-v1.6`	7,531	$+0.0022$	$[+0.0003, +0.0041]$
+ rich, `llava-med-v1.5`	7,531	$+0.0001$	$[-0.0018, +0.0020]^{\dagger}$
+ rich, `medgemma-4b`	7,531	$+0.0026$	$[+0.0005, +0.0048]$
+ rich, `medgemma-1.5`	7,531	-0.0003	$[-0.0025, +0.0019]^{\dagger}$
+ rich, `Lingshu-I-8B`	7,531	$+0.0032$	$[+0.0007, +0.0056]$
+ rich, `gemma-3-4b`	7,531	$+0.0012$	$[-0.0009, +0.0033]^{\dagger}$
+ rich, `gemma-4-E2B`	7,531	$+0.0048$	$[+0.0024, +0.0072]$

RadGraph-F1 on FINDINGS, paired percentile bootstrap over studies, $B = 10,000$, seed 0. $\hat{\theta}$ is the first-named arm minus the second. † marks an interval that covers zero. These intervals quantify test-set uncertainty only: every model was trained once, so they do not bound run-to-run variation.

Table 8: Every paired comparison the paper argues from. $\hat{\theta}$ is the first-named arm minus the second, on RadGraph-F1 over FINDINGS.

F Stratified context ablation

Strata are defined on the rendered prompt rather than on the source tables, so a stratum says what the model was actually told. A laboratory or microbiology summary counts as substantive when the summariser did not report that it found nothing significant. A study is infection-related when its reference labels carry Pneumonia or Consolidation. Each row carries its own paired interval, because a stratum is a smaller n and therefore a wider one. The full figures behind the compressed statements in Section 5.2 are here: the laboratory block is worth +0.002 $[+0.001, +0.003]$ where the summary is substantive against +0.001 $[-0.001, +0.003]$ where it is not, microbiology +0.002 against +0.001, and infection-related studies +0.001 on 587 studies against +0.002 elsewhere.

Why the null-manipulation control is not exactly zero. Adding priors to studies that have none is worth +0.001, and its interval excludes zero, at roughly a twenty-fifth of the effect it controls for (Section 6). For those studies the dropped block was empty to begin with, so a genuine effect is impossible. The likely cause is not a leak between tiers but the one deliberate asymmetry in the tier design. `baseline` is the only tier that receives an edited system prompt (Section 3, Appendix A), because the shared wording refers to prior studies that tier no longer supplies, so on a study with no priors the two tiers still differ in wording. Read that way, +0.001 is a measurement of that wording confound rather than a failure of the control, and it bounds it

Stratum	n	$\hat{\theta}$	95% CI
+ rich, substantive laboratory summary	6,455	+0.0019	[+0.0009, +0.0029]
+ rich, laboratory summary reports nothing significant	1,076	+0.0009	[−0.0013, +0.0031]†
+ rich, substantive microbiology summary	4,563	+0.0022	[+0.0011, +0.0033]
+ rich, microbiology summary reports nothing significant	2,968	+0.0012	[−0.0003, +0.0026]†
+ priors, studies that have priors	3,666	+0.0287	[+0.0251, +0.0326]
+ priors, studies that have none (null manipulation)	3,865	+0.0011	[+0.0002, +0.0019]
+ priors, infection-related studies	587	+0.0180	[+0.0100, +0.0270]
+ priors, all other studies	6,944	+0.0143	[+0.0123, +0.0163]
+ rich, infection-related studies	587	+0.0010	[−0.0016, +0.0036]†
+ rich, all other studies	6,944	+0.0018	[+0.0009, +0.0028]

RadGraph-F1 on FINDINGS, `reports` variant, pooled over the six backbones of the main comparison. Each row is one context-tier step restricted to the studies where that block did or did not carry content, as rendered into the prompt. † marks an interval that covers zero.

Table 9: The two context-tier steps restricted to the studies where each block did or did not carry content. The priors step is much larger where priors exist, as expected. The demographics/laboratory/microbiology step is *not* larger where those summaries are substantive, which is what rules out an averaging artifact as the explanation for its size (Section 5.2).

well below the effect it might contaminate.

Studies with priors are harder. Two further observations follow from the priors split. Studies with priors score 0.363 against 0.451 RadGraph-F1 at the richest tier, which is consistent with returning patients having more complex and more differential reports, and means the pooled priors effect is also diluted by the easier half of the cohort. And the small effect of laboratory and microbiology data is unaffected by the split: that block contributes about the same amount in both groups, so its weakness is not an artifact of averaging over studies it could not have helped.

G Metric availability by task

Metric	`find.`	`impr.`	`comb.`
BLEU	•	•	•
ROUGE-L	•	•	•
BERTScore	•	•	•
RadGraph-F1	•	•	•
F1-SRR-BERT	•	•	–
Category Score	•	–	–

find. = FINDINGS, *impr.* = IMPRESSION, *comb.* = the two concatenated. Category Score rolls FINDINGS organ categories up and is undefined elsewhere. F1-SRR-BERT is defined per report section and so is undefined on the concatenation. These are properties of the metrics, not scoring failures.

Table 10: Which metrics are defined on which task. The gaps are structural, not scoring failures: Category Score is defined over organ categories that only the FINDINGS section populates, and F1-SRR-BERT is undefined once FINDINGS and IMPRESSION are concatenated. This is why Table 3 carries all six metrics and Table 4 carries five.

H Per-label results

Every figure in this appendix is drawn over all seven backbones, including the one excluded from the main comparison (Appendix C). That is deliberate: these are diagnostic panels, and showing the excluded backbone alongside the others is part of what they are for. The numbers quoted in Sections 5 and 6 are computed over the six backbones of the main comparison, so a figure cell and a body number will not always agree on a family mean.

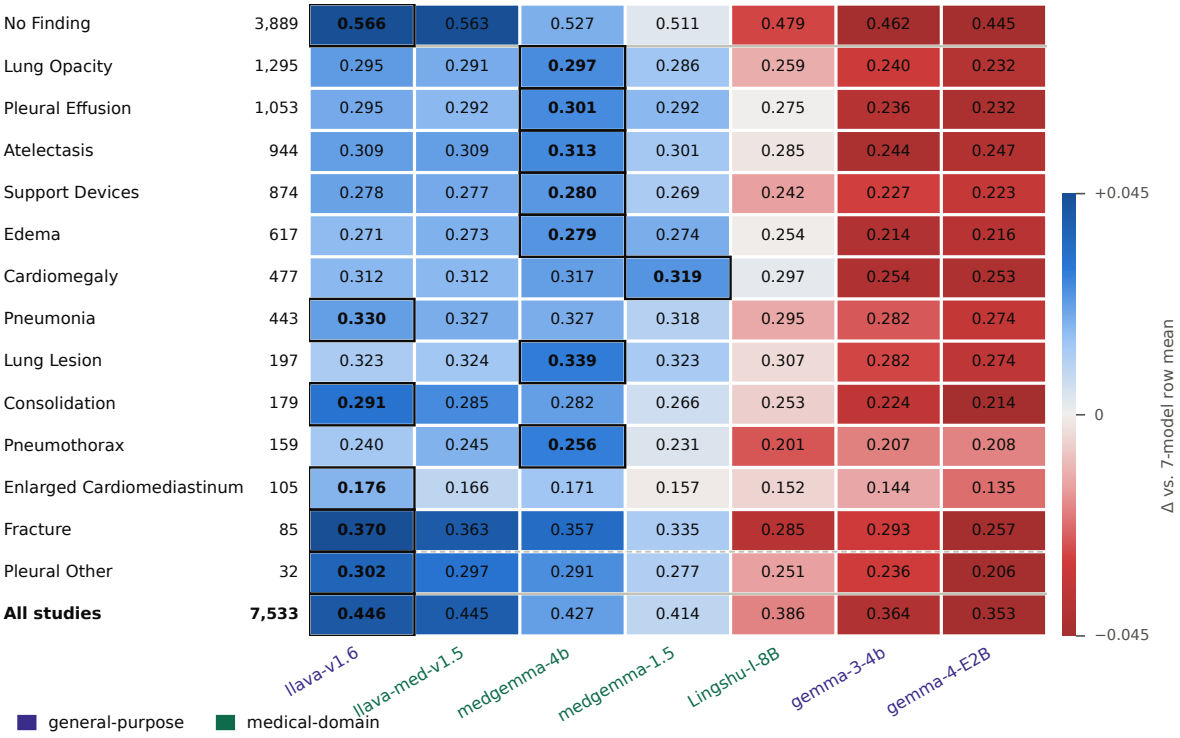


Figure 2: FINDINGS, positive labels: RadGraph-F1 restricted to studies positive for each CheXpert label (`reports` variant). Cells are shaded by deviation from the row mean, so the figure answers which model leads a label, not by how much. Labels below the dashed rule have fewer than 60 studies.

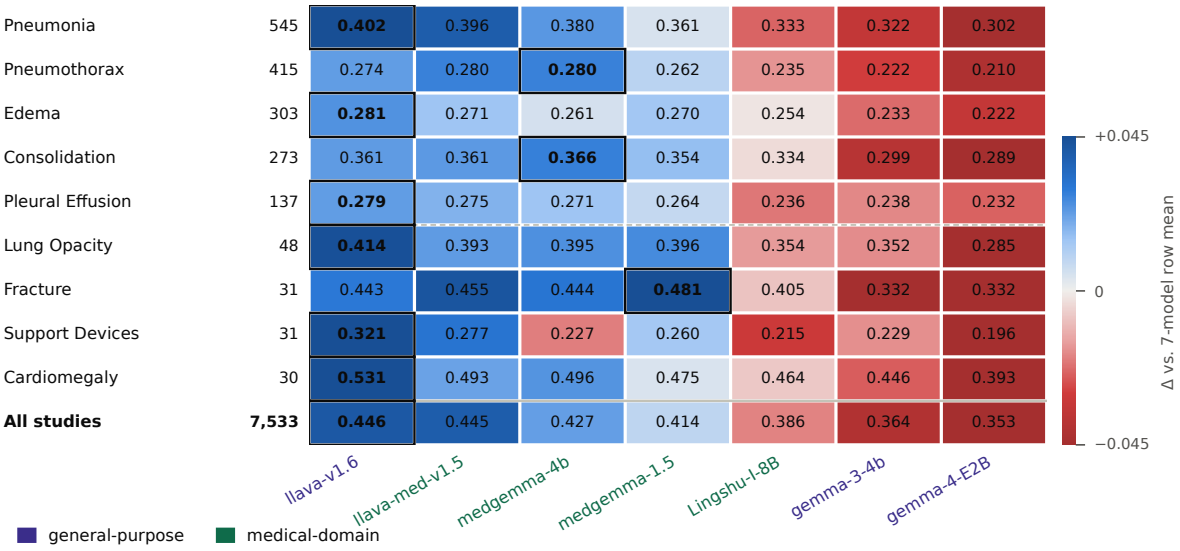


Figure 3: FINDINGS, negated labels: counterpart to Figure 2, restricted to studies where each label is explicitly negated.

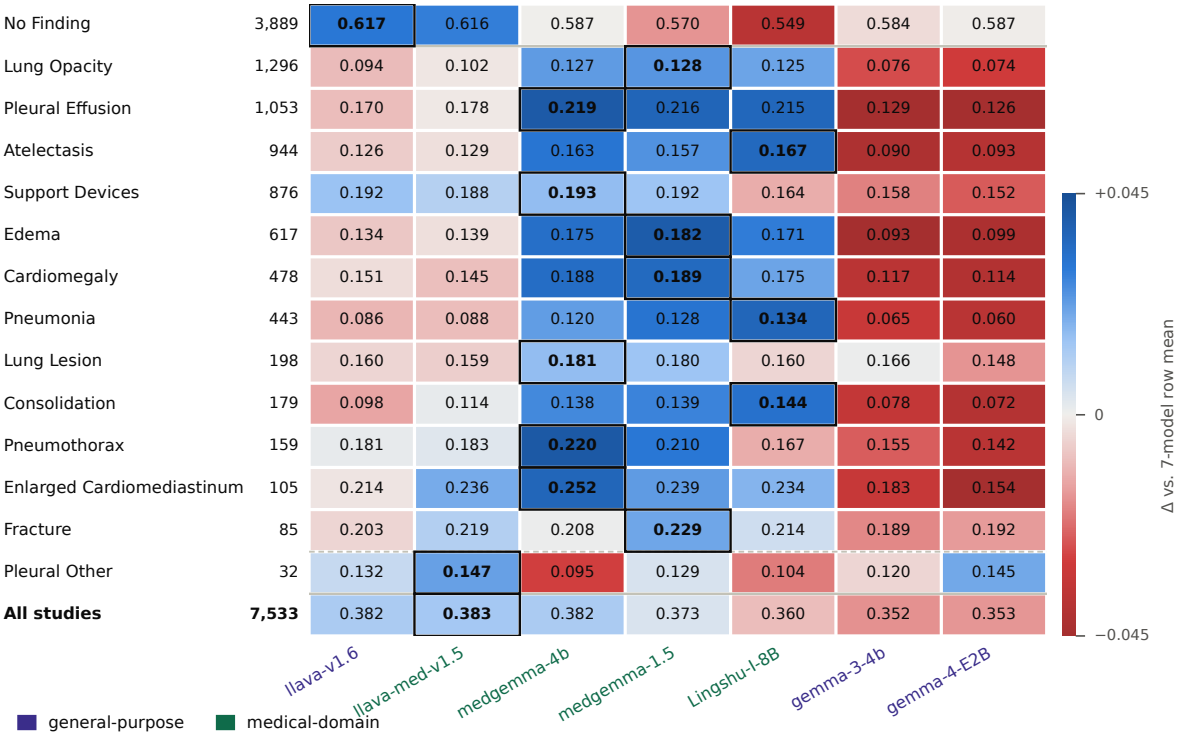


Figure 4: IMPRESSION, positive labels: counterpart to Figure 2.

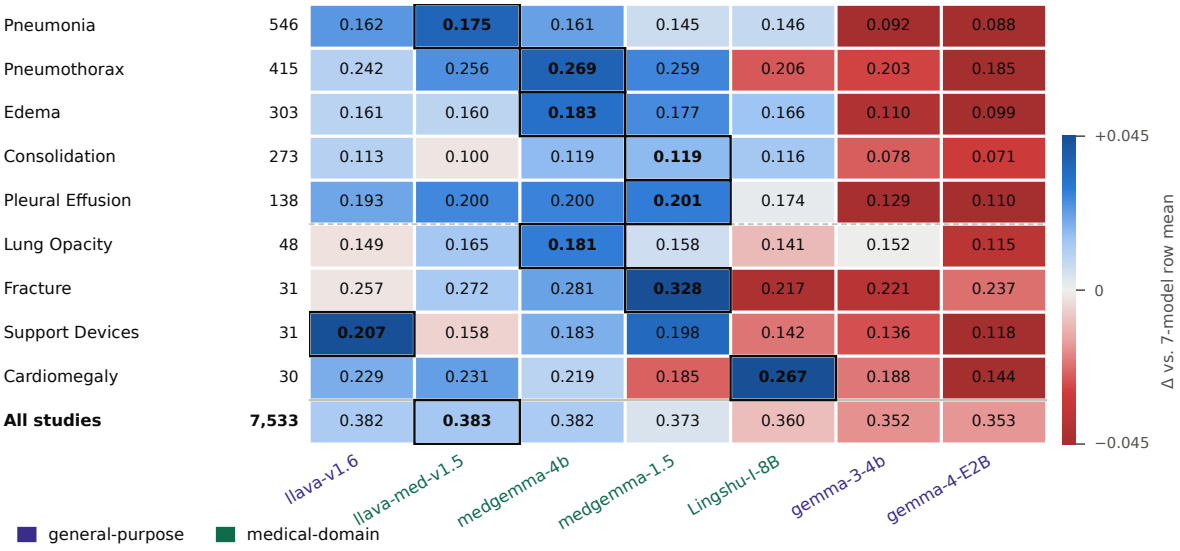


Figure 5: IMPRESSION, negated labels: counterpart to Figure 3. Per-cell scores span 0.071–0.383 against 0.196–0.531 on FINDINGS – IMPRESSION rarely restates a negation FINDINGS already made, leaving many cells with almost no extractable content. The all-studies row here is led by `llava-med-v1.5`, not `llava-v1.6` (Table 4), a property of the IMPRESSION task rather than of negation.

I Training variants and context tiers

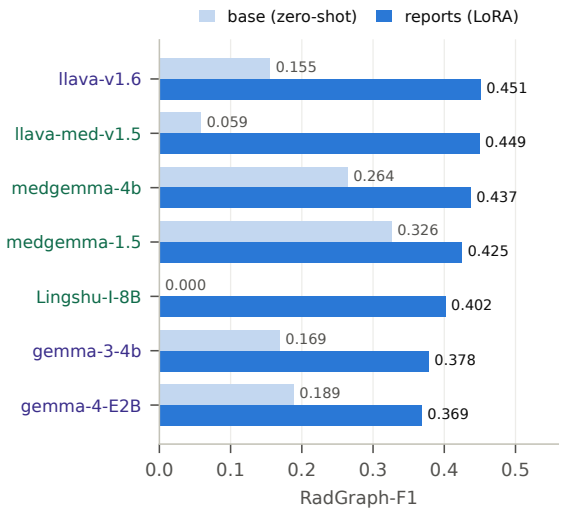


Figure 6: Zero-shot against fine-tuned RadGraph-F1 per backbone (`combined`), ordered by the fine-tuned score. The zero-shot bars are visibly not in that order (Section 5.1).

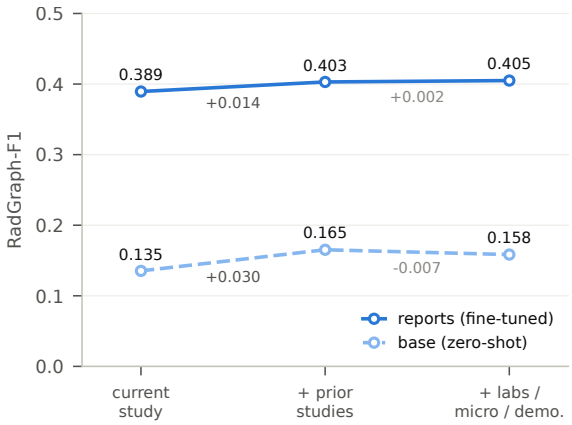


Figure 7: Context tiers per backbone. The step from *current* to + *priors* is visible, the step from + *priors* to + *rich* is not.

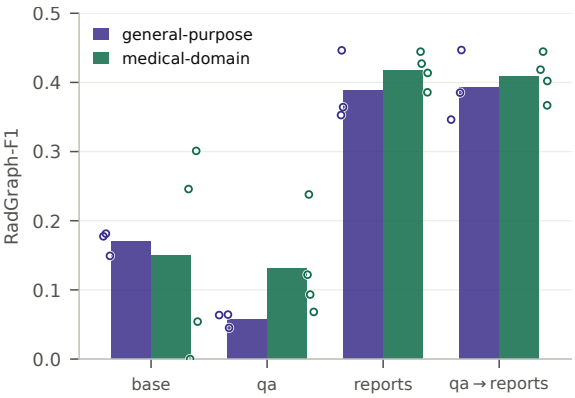


Figure 8: General and medical family means with per-backbone dispersion, FINDINGS. The dispersion within each family is larger than the gap between them.

	Context tier			Δ	
Model	current	+ priors	+ rich	priors	rich
`llava-v1.6`	0.428	0.444	0.446	+0.016	+0.002
`llava-med-v1.5`	0.428	0.444	0.445	+0.017	0.000
`medgemma-4b`	0.416	0.425	0.427	+0.009	+0.003
`medgemma-1.5`	0.401	0.414	0.414	+0.013	0.000
`Lingshu-I-8B`	0.375	0.383	0.386	+0.008	+0.003
`gemma-3-4b`	0.349	0.363	0.364	+0.014	+0.001
`gemma-4-E2B`	0.330	0.348	0.353	+0.018	+0.005
Mean	0.392	0.406	0.408	+0.015	+0.002

RadGraph-F1 on FINDINGS, `reports` variant. *current* = the study under report only, + *priors* adds prior reports, + *rich* adds demographics, laboratory results and microbiology. Tiers are literal block drops from one unchanged rendered prompt. The mean is over the six backbones of the main comparison. `Lingshu-I-8B` is listed for completeness but excluded from it.

Table 11: The three-tier context ablation, `reports` variant, FINDINGS. Tiers are literal block drops from one unchanged rendered prompt.

	Lexical			**Clinical**		
Model	BLEU	ROUGE-L	BERTScore	RadGraph-F1	F1-SRR-BERT	Cat. Score
`llava-v1.6`	**0.330**	**0.480**	**0.917**	**0.447**	**0.695**	**0.800**
`llava-med-v1.5`	0.325	0.478	0.916	0.445	0.689	0.793
`medgemma-4b`	0.293	0.451	0.913	0.418	0.682	0.787
`medgemma-1.5`	0.271	0.429	0.909	0.402	0.672	0.777
`Lingshu-I-8B`	0.259	0.403	0.904	0.367	0.665	0.786
`gemma-3-4b`	0.260	0.423	0.907	0.385	0.648	0.747
`gemma-4-E2B`	0.221	0.382	0.900	0.346	0.625	0.724

Table 12: Cross-model results on FINDINGS: all seven backbones, `qa_reports` variant, over the full test split of 7,533 studies. Best per column **underlined**.

	Lexical			**Clinical**	
Model	BLEU	ROUGE-L	BERTScore	RadGraph-F1	F1-SRR-BERT
`llava-v1.6`	**0.319**	**0.439**	0.908	**0.395**	**0.619**
`llava-med-v1.5`	0.314	0.433	0.909	0.389	0.613
`medgemma-4b`	0.297	0.420	**0.909**	0.376	0.613
`medgemma-1.5`	0.285	0.413	0.908	0.369	0.613
`Lingshu-I-8B`	0.272	0.396	0.905	0.352	0.596
`gemma-3-4b`	0.287	0.403	0.900	0.361	0.591
`gemma-4-E2B`	0.269	0.386	0.884	0.347	0.573

Table 13: Cross-model results on IMPRESSION: all seven backbones, `qa_reports` variant, over the full test split of 7,533 studies. Best per column **underlined**.

	Report text (FINDINGS and IMPRESSION)
Reference	FINDINGS: In comparison to the prior examination from a few hours prior there has been interval placement of an enteric tube which courses below the level of the diaphragm and likely terminates in the region of the stomach. There has also been interval placement of right IJ line which ends in the upper SVC. Moderate edema is worsened. Probable small pleural effusions. IMPRESSION: An enteric tube courses below below the diaphragm and terminates in the region of the stomach. A right IJ line ends in the upper SVC.Moderate edema is worsened.
`base`	IMPRESSION: 1. Mild to moderate pulmonary edema, which appears to be stable compared to the previous exam. 2. Small bilateral pleural effusions, which have decreased in size. 3. Persistent peripheral wedge-shaped opacity within the right lower lobe, which may represent an area of infarction or persistent infection. 4. Other previously described multifocal airspace opacities within the right lung and left infrahilar region have improved, suggesting areas of improving infection. FINDINGS: * Cardiac silhouette size remains mildly enlarged. * The mediastinal contour is unchanged. * Perihilar haziness and vascular indistinctness is compatible with mild pulmonary edema, which has improved slightly. * No large pleural effusion or pneumothorax is seen. * Previously noted peripheral opacity in the right lung base continues to improve. * Atelectatic changes are re-demonstrated in the lung bases. * No acute osseous abnormality is visualized. CLINICAL CONTEXT: The patient is an 86-year-old female with a history of shortness of breath and congestive heart failure. She has exhibited significant metabolic and hematologic abnormalities, including acidemia, electrolyte imbalances, renal function decline, anemia, and coagulation abnormalities. Multiple cultures have been performed to monitor for infection following central line, NG, and ET tube placement. The most recent urine culture identified Escherichia coli, which was sensitive to multiple antibiotics including Nitrofurantoin, Cefazolin, and Ciprofloxacin, alongside Gram-positive bacteria. Subsequent urine cultures showed no growth or low organism counts. Legionella urinary antigen was negative, and viral culture was negative. A urine culture on 2135-09-11 showed mixed bacterial flora consistent with contamination. Stool testing for C. difficile PCR was negative, and glucose levels were elevated.
`qa`	The cardiac silhouette is unchanged.
`reports`	FINDINGS: The endotracheal tube terminates 2.5 cm above the carina. The nasogastric tube terminates in the stomach. The right internal jugular venous catheter terminates in the mid SVC. The lung volume is small. The left lower lobe atelectasis is improved. The left pleural effusion is unchanged. The right pleural effusion is small. The pulmonary vessel congestion is mild. The cardiac silhouette is enlarged. The mediastinum is unchanged. IMPRESSION: 1. The endotracheal tube terminates 2.5 cm above the carina. 2. The nasogastric tube terminates in the stomach. 3. The pulmonary vessel congestion is mild.
`qa_reports`	FINDINGS: The endotracheal tube terminates 3.5 cm above the carina. A nasogastric tube terminates in the stomach. A right internal jugular central venous catheter terminates in the mid SVC. There is no pneumothorax. There is no focal consolidation or pleural effusion. Mild pulmonary edema is unchanged. IMPRESSION: 1. Appropriate positioning of the endotracheal tube and nasogastric tube. 2. Stable mild pulmonary edema.

Table 14: One test study through `llava-v1.6`, all four variants, verbatim and untruncated. `base` inverts the two sections, writes markdown, reverses the reference's direction of change, and runs on into a `CLINICAL CONTEXT` section echoing the prompt back. `qa` answers in one sentence with no sections, which is the VQA format rather than a report and is what Section 6 argues Stage A teaches. `reports` and `qa_reports` produce the two requested sections and recover both devices, but understate the worsened moderate edema.

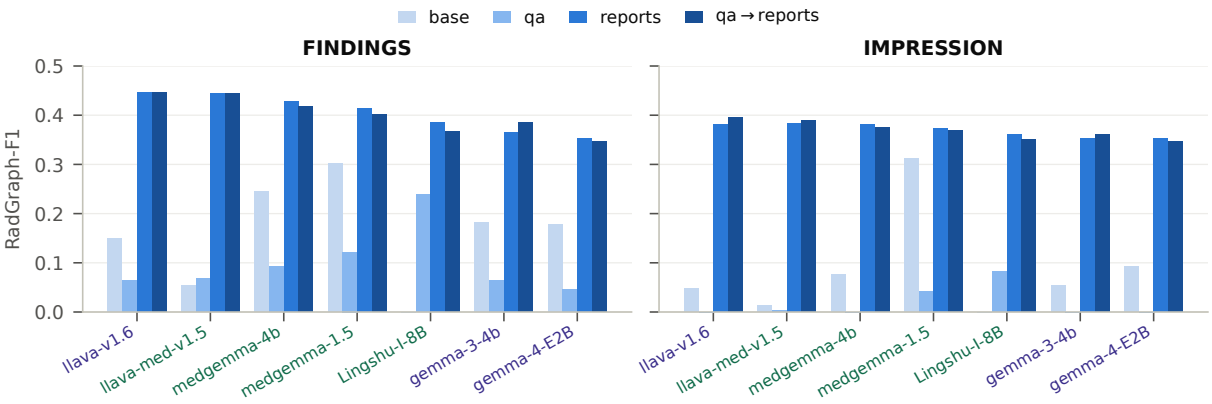


Figure 9: All four training variants, by report section, over all seven backbones. The `qa` variant scores below the untrained `base` checkpoint on 5 of 6 backbones of the main comparison. `qa_reports` moves `reports` by a small, backbone-dependent amount whose sign is not consistent (Section 5.1, Table 8).

J Metric package versions

BLEU: `nltk` 3.9.4 (`corpus_bleu`). ROUGE-L: `rouge-score` 0.1.2. BERTScore: `bert-score` 0.3.13. RadGraph-F1: `radgraph` 0.1.18. F1-SRR-BERT and Category Score: `StanfordAIMI/SRR-BERT-Leaves`, loaded by model ID with no pinned revision. All other package versions are in `uv.lock` at the repository root.